

**a**

F1a Pearson, three readings

held-out Pearson

ladder to  $\lambda = 100$ ;  
random at  $\lambda = 1$  (and 0.1)  
replaces random at 1e-3  
stars per reading

log10  $\lambda$ , none, mask, rand

**b**

F1b held-out loss

point loss

epoch 29 and  
at min val loss

log10  $\lambda$

**c**

F1c edge recall

recall at degree

vs own target;  
random arms also  
vs the biological  
graphs (from ckpt)

log10  $\lambda$

**d**

F1d divergence

divergence

floor at ~300:  
does 10, 100  
break it?

log10  $\lambda$

**e**

F1e gradient ratio

penalty / point

adds 10, 100  
and random 1

probe epoch

**f**

F1f random control

Pearson, ep 29

biological vs random  
vs none, AT  $\lambda = 1$   
(n = 3 each)

target

**g**

F2a Pearson, three readings

held-out Pearson

one column per mechanism;  
stars vs none

mechanism, reach and direction (9 columns)

**h**

F2b held-out loss

point loss

epoch 29 and  
at min val loss

mechanism

**i**

F2c edge recall

recall at degree

mask arms 1 by  
construction on their  
own support; score all  
vs one-hop biological

mechanism

**j**

F2d val Pearson by epoch

Pearson

none, mask 1/2/3-hop,  
KL 1 1/2/3-hop  
(the curves move here  
from F1 g)

epoch

**k**

F2e train Pearson by epoch

Pearson

same arms

epoch

**l**

F2f val loss by epoch

point loss

same arms

epoch

**m**

F3a 60 epochs, val Pearson

Pearson

none, mask, KL 1;  
does KL 1 keep rising,  
does none keep falling

epoch (0-59)

**n**

F3b placement curves

Pearson

val Pearson by epoch  
for the placement arms

epoch

**o**

F3c discovery

recall of unseen edges

from best checkpoints:  
regulatory head vs  
TFLink-only edges;  
null: degree-matched

top-k off-graph attention

**p**

F3d overruling

share with  $|e| > 0.08$

from best checkpoints:  
kept vs down-weighted  
prior edges against  
Costanzo digenic  $\epsilon$

$\alpha_{ij} / (1/d_{ij})$ , binned

**q**

F3e divergence, all arms

divergence to biological  $\tilde{\lambda}$

none, mask, random,  
symmetric: from ckpt,  
fills the crosses of F1 d

arm

**r**

F3f seed table

per-run table of every  
round-2 run, W&B ids,  
as t2 today